Nathaniel Pritchard

October 2026

CONTACT

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RESEARCH INTERESTS

Randomized Numerical Linear Algebra. Scientific Computing. Statistical Computation. Generalized Linear Models. High Performance Computing. Bayesian Statistics. Iterative Solvers. Optimization. Preconditioners.

EDUCATION

University of Wisconsin-Madison, Madison, WI
Ph.D., Statistics August 2018 - August 2024
Adviser: Vivak Patel

University of North Carolina at Chapel Hill Chapel Hill, NC
B.S., Statistics and Analytics August 2014 - May 2018
(Highest Honors and Highest Distinction)
Honors Thesis Adviser: Shankar Bhamidi

RESEARCH EMPLOYMENT

University of Oxford, Oxford, United Kingdom
Postdoctoral Research Associate October 2024 -
Supervisor: Yuji Nakasukasa and Per-Gunnar Martinsson
Topic: Certifying accuracy of randomized algorithms in numerical linear algebra

University of Wisconsin-Madison, Madison, WI
Research Assistant August 2023 - May 2024
Supervisor: Vivak Patel
Topic: Development of Randomized Linear Algebra Library for Julia

Argonne National Laboratory, Chicago, IL
Givens Associate May 2023 - July 2023
Supervisor: Adrian Maldonado
Topic: Accelerating Newton-Raphson on GPUs using deflation methods

Argonne National Laboratory, Chicago, IL
Givens Associate May 2020 - July 2020
Supervisor: Adrian Maldonado
Topic: Preconditioners for solving graph Laplacians arising from power grid networks

PRE-PRINTS

Lazzarino, L., Pearce, K., & **Pritchard, N.** (2026). Efficient error estimators for Generalized Nyström. arXiv preprint arXiv:2601.11493.

Pritchard, N., Park, T., Nakasukasa, Y., & Martinsson, G. (2025). Fast Rank Adaptive CUR via a Recycled Small Sketch. arXiv preprint arXiv:2509.21963.

Patel, V., Maldonado, D. A., Melnichenko, M., **Pritchard, N.**, Rao, V., Rebrova, E., & Sankararaman, S. (2025). Scientific Applications Leveraging Randomized Linear Algebra. arXiv preprint arXiv:2506.16457.

PUBLICATIONS

Pritchard, N., & Patel, V. (2023). Solving, Tracking and Stopping Streaming Linear Inverse Problems. *Inverse Problems*, 40 (8), 10.1088/1361-6420/ad5583.

Pritchard, N., & Patel, V. (2023). Towards Practical Large-Scale Randomized Iterative Least Squares Solvers through Uncertainty Quantification. *SIAM/ASA Journal on Uncertainty Quantification*, 11(3), 10.1137/22M1515057.

He, M., Glasser, J., **Pritchard, N.**, Bhamidi, S., & Kaza, N. (2020). Demarcating geographic regions using community detection in commuting networks with significant self-loops. *PLOS ONE* 15(4), e0230941.

TALKS

"o(n) Eigensolvers" September 2026 ENLARGED at Rutherford Appelton Lab in Didcot, Uk.

"RandLinearAlgebra.jl: A cyberlaboratory for RandNLA" June 2026 RandNLA workshop in Lausanne, Switzerland.

"IterativeCUR: One Small Sketch for Big Matrix Approximations" May 2026 International Linear Algebra Society in Blacksburg Virginia.

"RandLinearAlgebra.jl" May 2026 at Midwest Randomized Linear Algebra Workshop in Madison, WI.

"IterativeCUR: One Small Sketch for Big Matrix Approximations" February 2026 at Numerical Analysis Internal Seminar in Oxford, UK.

"IterativeCUR: One Small Sketch for Big Matrix Approximations" November 2025 Applied and Computational Analysis (ACA) seminar in Cambridge, UK.

"Small Sketches for Big Matrix Approximations" September 2025 ENUMATH in Heidelberg, Germany.

"Small Sketches for Big Matrix Approximations" July 2025 SIAM Annual Meeting in Montreal, Canada.

"Small Sketches for Big Matrix Approximations" June 2025 at The 30th Biennial Numerical Analysis Conference in Glasgow, Scotland.

"Statistical Approaches to Numerical Linear Algebra" March 2025 at Optimization and Numerical Linear Algebra Exposé at the Rutherford Appleton Laboratory in Didcot, UK.

"Who needs a residual when an approximation will do?" December 2024 at

Numerical Analysis Internal Seminar in Oxford, UK.

"Tracking Progress in a Big Data World" October 2024 at SIAM Mathematics of Data Science in Atlanta, Georgia.

"Iterative Right Random Sketching for Large-Scale Logistic Regression" August 2024 at Joint Statistical Meetings in Portland, Oregon.

"Computationally Efficient Tracking for Iterative Random Sketching" May 2024 at SIAM Applied Linear Algebra Conference in Paris, France.

"Computationally Efficient Tracking for Iterative Random Sketching" April 2024 at IFDS Ideas Forum in Madison, Wisconsin.

"Large-scale randomized iterative least squares" March 2023 at SIAM CSE in Amsterdam, Netherlands.

"Towards practical large-scale least squares solvers with Iterative Right Random Sketching" February 2023 at Argonne National Labs LANS seminar in Chicago, Illinois (Virtual).

"Residual Tracking and Stopping for Iterative Random Sketching" April 2022 at Copper Mountain Conference on Iterative Methods in Copper Mountain, Colorado (Virtual).

HONORS & AWARDS

SIAM Student Travel Award (SIAM LA 2024) May 2024.

Outstanding TA May 2023.

Student Research Grant Competition UW - Madison February 2023.

Outstanding TA (Honorable Mention) May 2022.

TEACHING EXPERIENCE

Teaching Assistant (Undergraduate): Introductory Applied Statistics for the Life Sciences (Fall 2018 - Spring 2019), Introduction to Statistical Methods (Fall 2019), Introduction to Statistics for Science and Engineering (Spring 2019), Data Science Modeling I (Fall 2020 - Spring 2022, Spring 2023)
Teaching Assistant (Graduate): Statistical Methods I (Fall 2022)
Instructor of Record: Data Science Modeling I (Spring 2023)*
*Three weeks into this semester, the instructor of this course had to take leave. In addition to my regular TA duties, I was put in charge of all the course logistics with regard to managing assignments, the course website, student questions, exams, and grades.

SUPERVISION

Master's: Aleksander Markiewicz (*Operator Learning for Near Symmetric Matrices*)

SERVICE

Organizer: Numerical Analysis Analysis Group Internal Seminar (2025 - 2026), North-South Colloquium (2025 - 2026)

Committee Service: TA Training Redesign (2020), Space Committee (2021), Awards and Outreach Committee (2022), Postdoc representative for Health and Safety Committee (2025-2026), Postdoc head of Fridays at 4 (2025-2026)

Statistics Graduate Student Association: Outreach Chair (Spring 2018 - Fall 2020), President (Fall 2020 - Spring 2023)

Journal Reviewer: IMA Journal of Numerical Analysis, SIAM Journal of Scientific Computing, IEEE Signal Processing Letters

SKILLS

Computer Languages: Julia, R, C, C++, Python

APIs: CUDA, MPI, OpenMP, PETSc

SOFTWARE

RandLinearAlgebra (Lead Developer): A Julia Package for prototyping and benchmarking Randomized Linear Algebra algorithms.